



**Customer Loyalty Analysis at Turtle Games Using Sentiment Analysis,  
Natural Language Processing and Linear Regression Modelling in Python  
and R**

LSE DA301

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## 1. Background and context

Using the 5 why's Framework (Toyoda 1993) a business problem statement was created to assist Turtle Games improve their overall sales performance. Turtle Games, a game development company has been struggling to understand what motivates their shoppers to accumulate loyalty points and the different consumption habits between demographics. To remain competitive, Turtle Games should employ a data driven approach informed by review sentiment analysis and linear regression modelling on customer shopping and review data to guide their strategic marketing efforts. This analysis will investigate Turtle Games' current plan using three business objectives:

|                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Determine the key drivers of loyalty point accumulation by applying linear regression and Random Forest modeling to evaluate the influence of remuneration, age, and spending score. |
| Segment the customer base into distinct groups using K-means clustering to identify high-value demographics and under-utilised marketing opportunities.                              |
| Gauge customer opinion and product performance through sentiment analysis of customer reviews to pinpoint specific product improvements and guide targeted marketing campaigns.      |

## 1. Analytical approach

This analysis was performed using both programming languages: Python and R. Initial investigation and cleaning techniques were performed in Python. Within Python, the 'language' and 'platform' columns were dropped due to redundancy and column names were changed for ease of use (Appendix 1.1).

The data was further cleaned by checking for duplicates and outliers. After cleaning was complete, analysis began with linear regression modelling using the 'sklearn' library to grasp initial insight into predictors of loyalty points. The predictors tested were the numeric variables: remuneration, spending score and age (Appendix 1.2 - 1.4). Initial regression modelling alluded to both spending score and remuneration as indicators of loyalty points. Age was not initially proven to be an indicator as  $R^2$  value was -0.04 and a pvalue of above 0.05, leading us to reject  $H_0$  (age as a key indicator of loyalty points). A decision tree was also created using sklearn to further investigate trends in the data. Here spending score came out as they key indicator of loyalty points ( Appendix 1.5).

Further analysis investigated groups within the dataset (specifically remuneration and spending score) to help Turtle Games understand their customer base. Kmeans was used to perform clustering analysis where 5 was determined to be the best fit for cluster size (Appendix 1.6-1.9).

Further analysis investigated customer review sentiment in both Python and R. In Python, the text within the 'review' and 'summary' columns were cleaned for analysis (Appendix 1.10) and then checked for frequency distribution and sentiment scores. Overall checks showed that sentiment across the board skewed positive, with some specific product based cases where sentiment compounds skewed low (Appendix 1.11)

Further linear analysis was performed using R and the tidyverse library. The moments package was used to test for skew and kurtosis and a shapiro wilkes test was performed to test for normality (Appendix 1.13). Next, a relationship pairplot was created to investigate correlation between variables and their influence on loyalty point accumulation. Beyond this a MLR (Multiple Linear Regression) model and a Random Forest Model were created to predict potential future loyalty point accumulation based on age, remuneration and spending score (appendix 1.14-1.15). The Random Forest model was split into an 80/20 train/ test split to avoid biased predictive data within the model.

## **2. Visualisation and insights**

To address our business objectives, four main visualisations have been selected to help guide Turtle Games understand brand positioning and strategy.

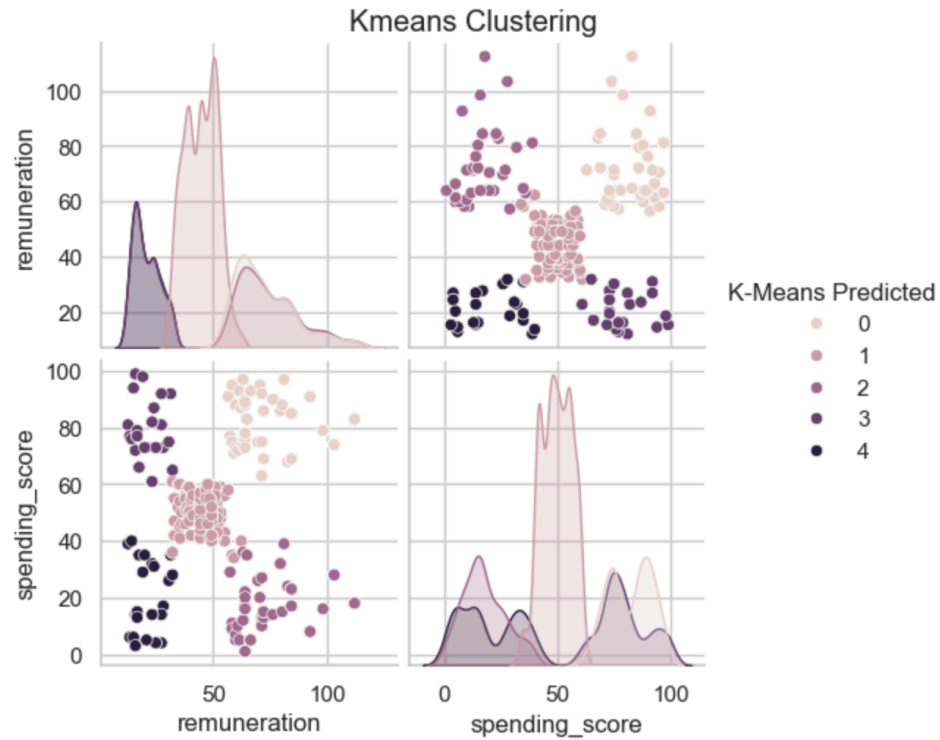
## 1. Relationship and correlation pair plot.

Relationship Pairplot

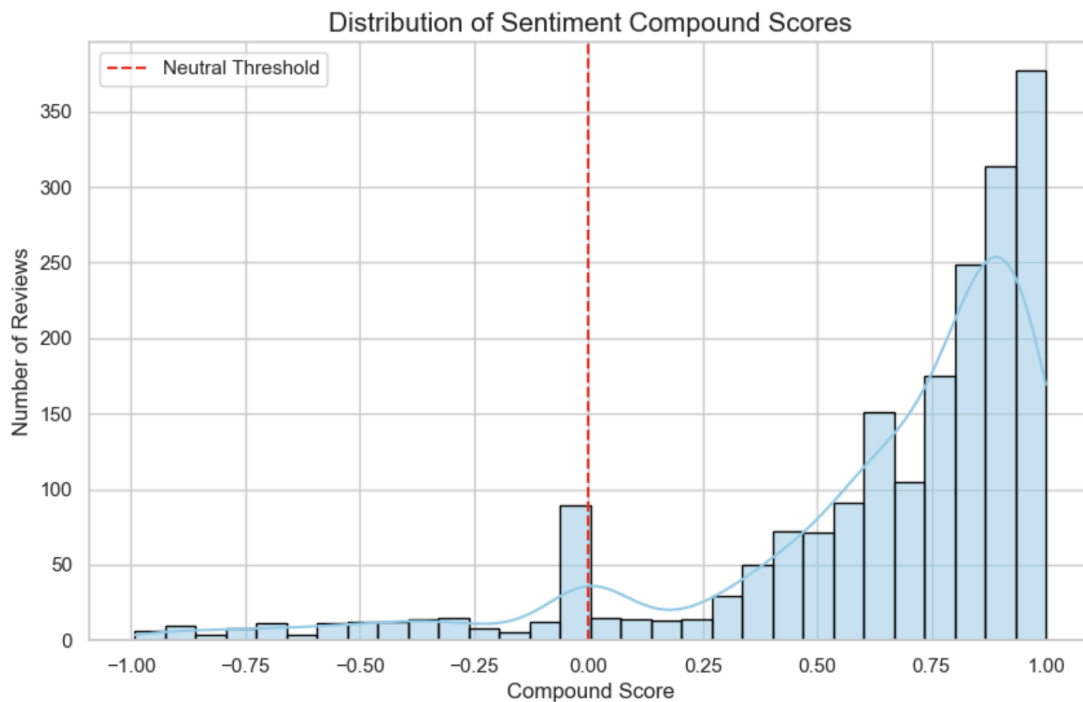


This plot represents the distribution and interaction of variables 'age', 'remuneration', 'spending score' and 'loyalty points'. This visualisation clearly shows correlation between variables and highlights the significant values with asterisks. We can see the strongest correlation in loyalty point prediction is spending score ( $R^2 = 0.67$ ) followed closely by remuneration ( $R^2 = 0.62$ ). It also shows us distinct 5 groups within remuneration versus spending score. This visualisation justified the use of a MLR model as we can clearly see that Loyalty points are not randomly distributed and follow specific patterns.

## 2. Customer Segmentation using Kmeans clustering



### 3. Customer Review Sentiment Distribution



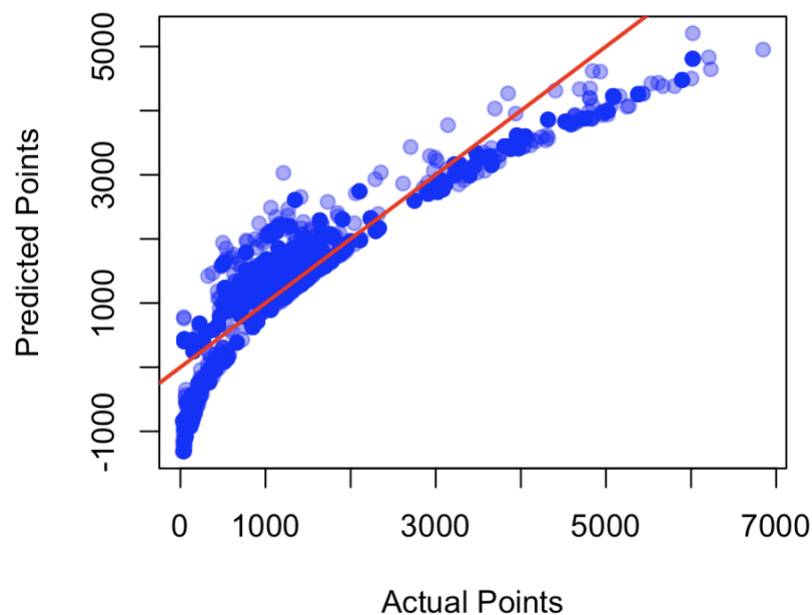
A histogram of compound sentiment scores was created to evaluate review data. The distribution was heavily positively skewed (0.5 to 1.0 range), providing evidence that

Turtle Games customers are generally happy with the product selection. However, by isolating the tail of the distribution, we identified specific product IDs (9090, 263) that were outliers in customer satisfaction.

|         | neg      | neu      | pos      | compound |
|---------|----------|----------|----------|----------|
| product |          |          |          |          |
| 9080.0  | 0.115333 | 0.669444 | 0.215444 | 0.281211 |
| 263.0   | 0.094000 | 0.647333 | 0.258500 | 0.283667 |
| 1241.0  | 0.061700 | 0.668300 | 0.270000 | 0.303040 |
| 3427.0  | 0.081900 | 0.708300 | 0.209700 | 0.330350 |
| 3711.0  | 0.046375 | 0.751750 | 0.202000 | 0.336750 |
| 4452.0  | 0.044000 | 0.776778 | 0.179333 | 0.353756 |
| 466.0   | 0.097800 | 0.765600 | 0.136800 | 0.364060 |
| 7573.0  | 0.022100 | 0.798500 | 0.179400 | 0.375100 |
| 6271.0  | 0.026222 | 0.794667 | 0.179111 | 0.393911 |
| 10196.0 | 0.029800 | 0.653300 | 0.316900 | 0.409140 |

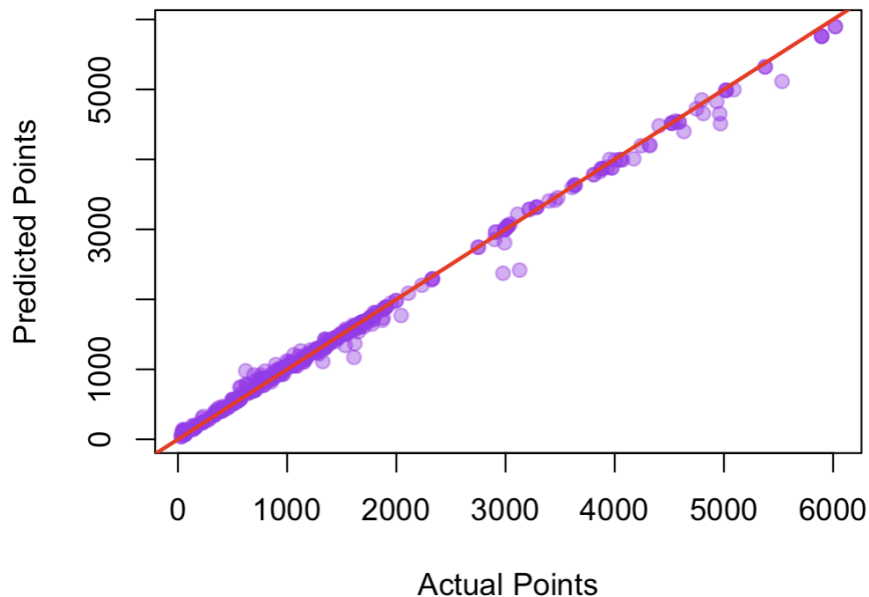
#### 4.1 MLR actual vs predicted plot

##### Actual vs. Predicted Loyalty Points



#### 4.2 Random Forest Actual vs Predicted Plot

### Random Forest: Actual vs. Predicted (Test Set)



MLR(multiple linear regression) and Random forest models were created to predict possible future purchase data. These were then plotted to visualise the actual vs predicted scores. The initial MLR model has slight curvature which justified the implementation of a Random forest model. The Random forest model was split into 80% train and 20% test to feed the model with unbiased data. The random forest model had an  $R^2$  value of 0.995 and was neatly centered around the 45 degree line, proving very accurate predictive capabilities. While the shapiro wilkes test confirmed a lack of normality across each variable, the random forest model proved to be the best fit in predicting loyalty points from a customer's age, spending score and remuneration.

#### 4. Patterns and predictions

The most poignant pattern identified in the analysis was the relationship between spending score and remuneration on loyalty point accumulation, specifically the high correlation between spending score and loyalty points ( $R^2 = 0.67$ ) closely followed by remuneration ( $R^2 = 0.62$ ), which suggests that the Turtle Games Loyalty points programme is centered around how much a customer spends on products rather than how often they purchase. Upon further inspection on the trends within the spending score variable, it was determined that customers aged 25-35 have the highest average spending scores with a steady decline after the age of 35 (Appendix 2.1). This positions

the 25-35 demographic as the most valuable in terms of loyalty point accumulation and should be the target for marketing efforts moving forward.

To predict future loyalty point accumulation, Turtle Games should use the Random Forest model due to its strong  $R^2$  value attained after being split into train/test to avoid overfitting. Spending score remained the dominant predictor of loyalty points. Based on these findings, the following recommendations can be made:

|                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Customer targeting: marketing efforts should focus on high earning individuals between 25-35 who represent the demographic with the highest spending score and largest potential for loyalty point accumulation |
| Customer conversion: marketing strategy should focus on high earning and low spending customers (cluster 4) and introduce premium product categories to potentially increase spending frequency.                |
| Product review: Quality control should be implemented for products 9080 and 263 which have significantly lower compound sentiment scores ( $<0.3$ ) compared to the average.                                    |



## Appendix

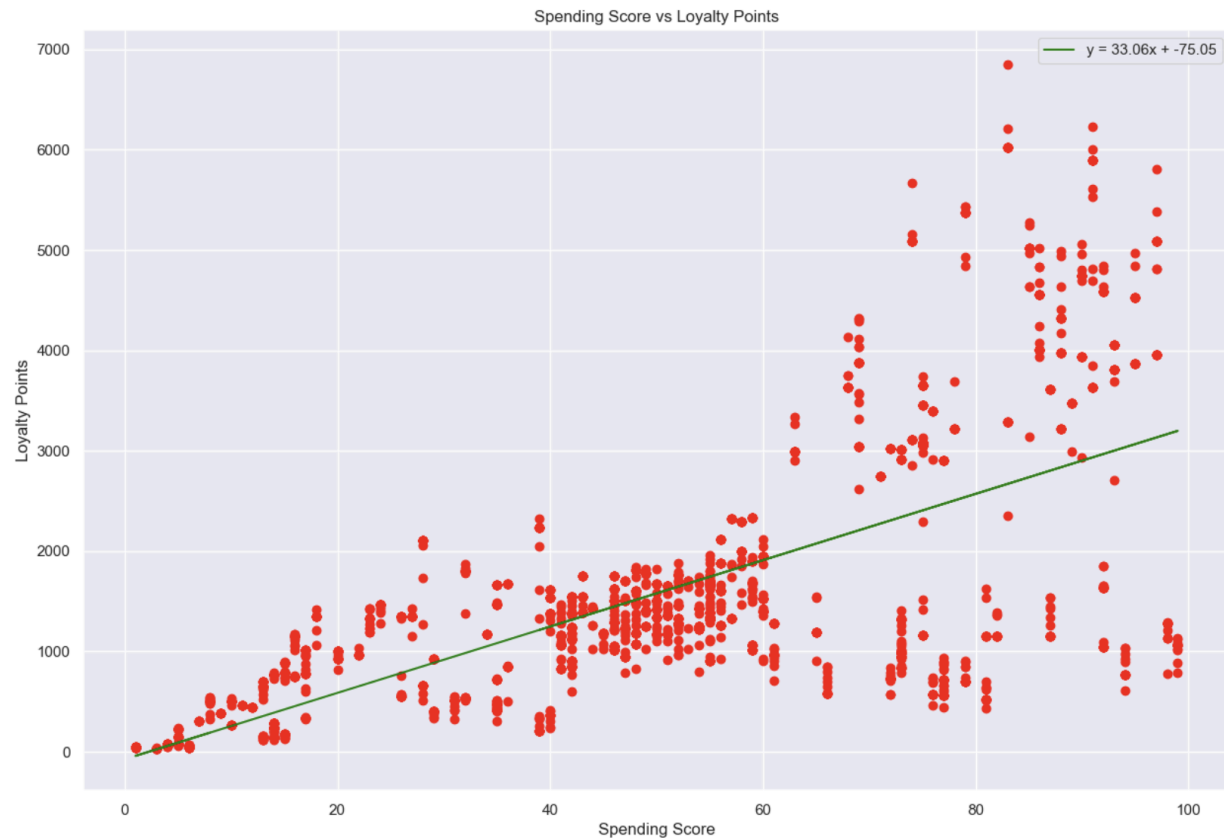
### 1.1.

```
# renaming columns
reviews = reviews.rename(columns={'remuneration (k£)' : 'remuneration',
                                  'spending_score (1-100)' : 'spending_score'})

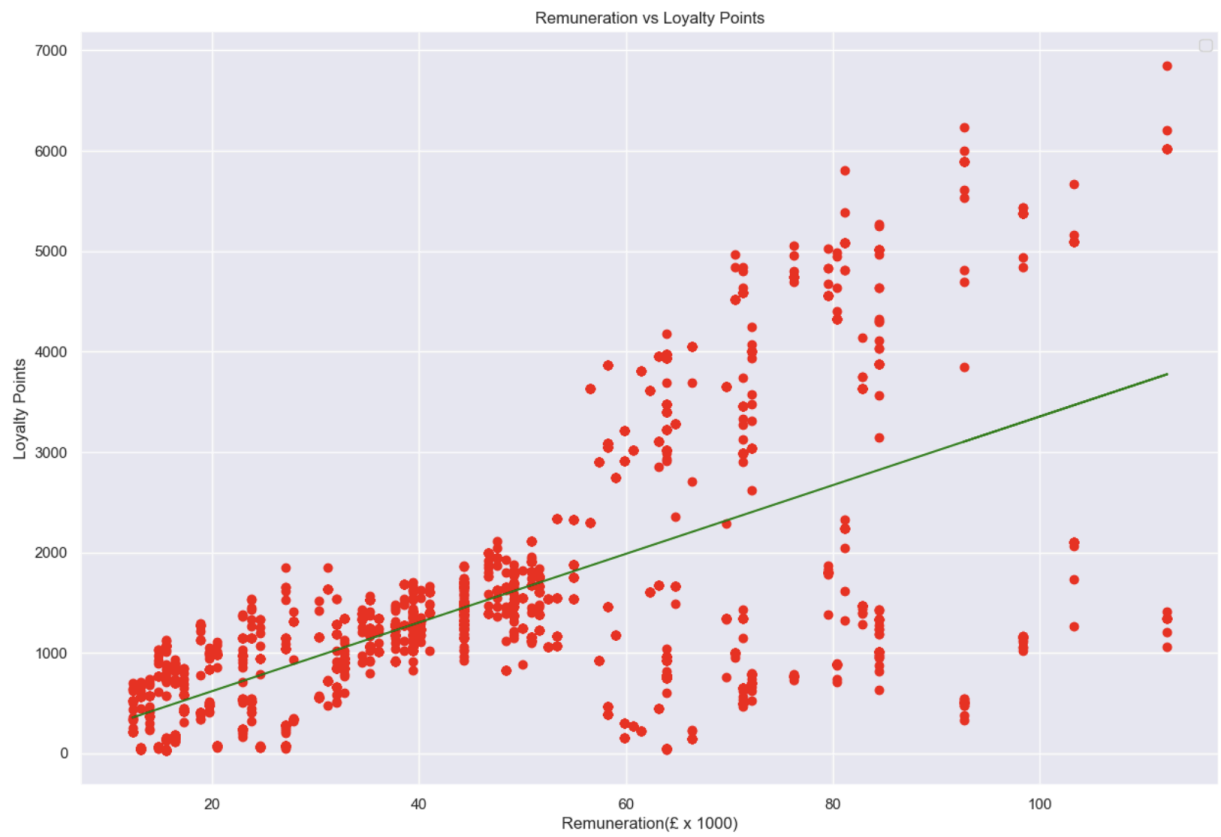
# Drop unnecessary columns.
reviews = reviews.drop(columns=['language', 'platform'])
```

Column name changes. 'remuneration (k£)' was changed to 'remuneration' and 'spending\_score (1-100)' was changed to 'spending score'

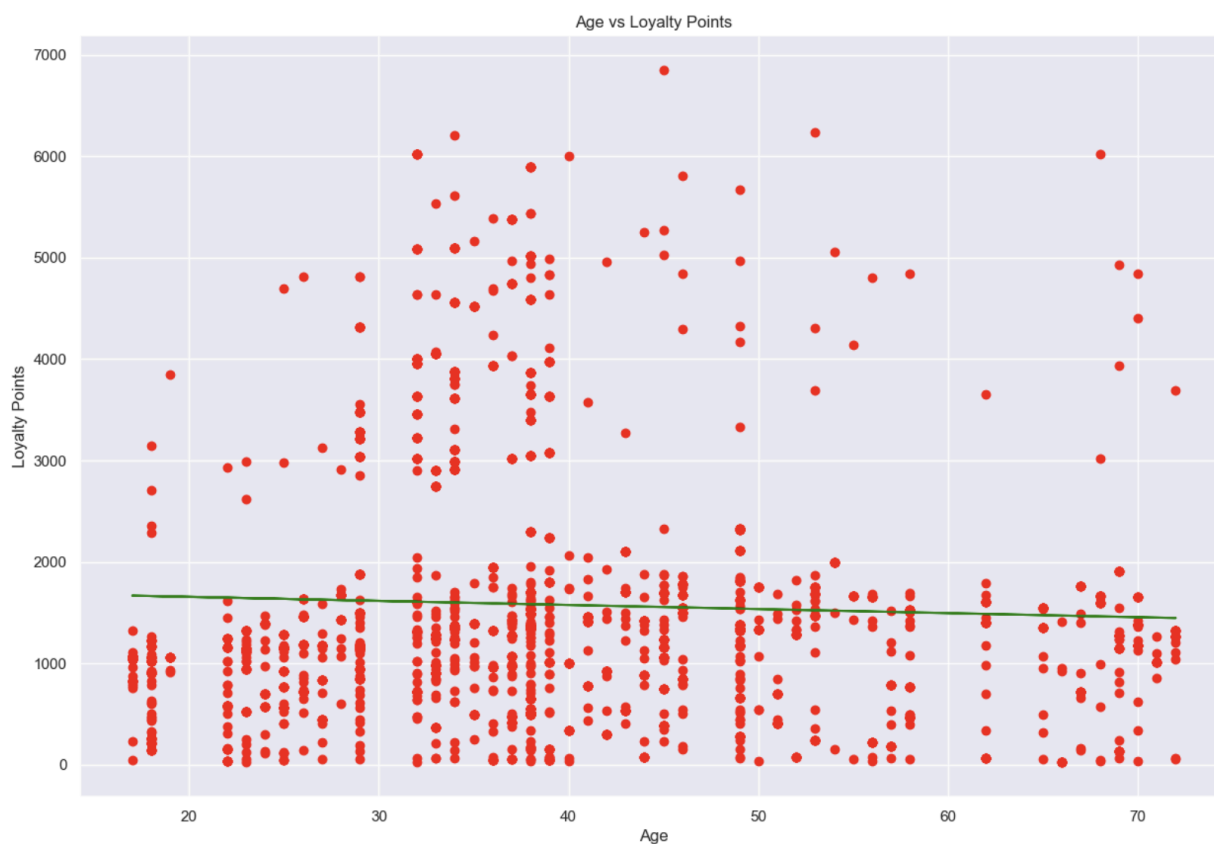
### 1.2 Linear Regression modelling results using Python: **Spending Score vs Loyalty points**



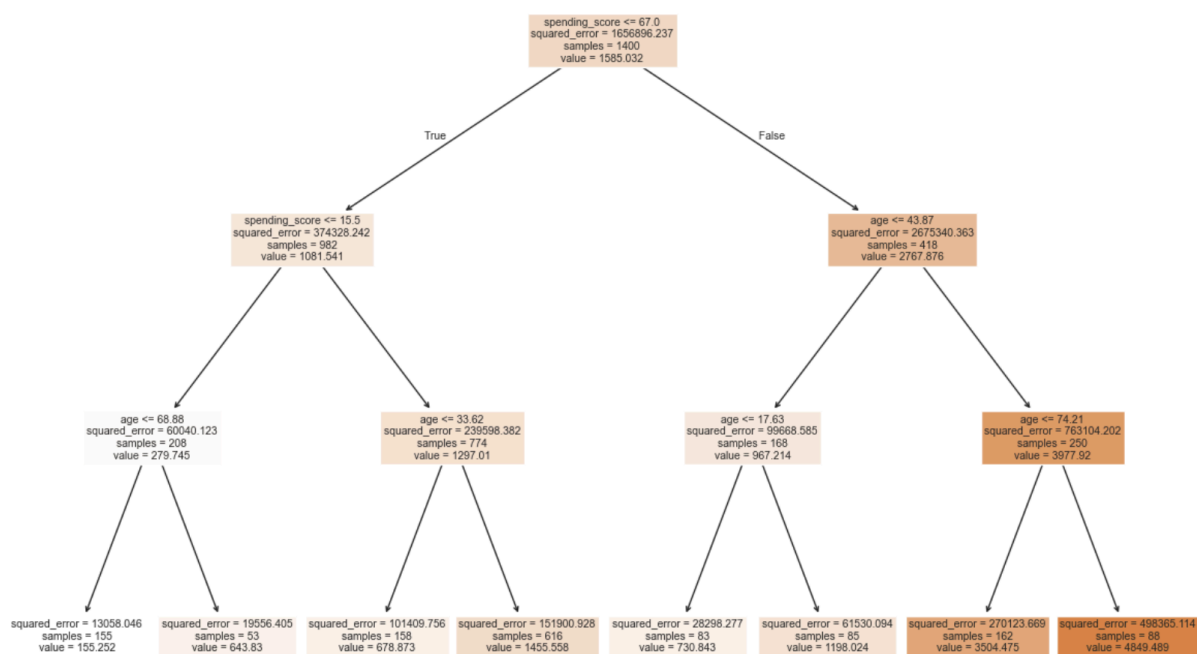
### 1.3 Linear Regression modelling results using Python: **Remuneration vs Loyalty points**



## 1.4 Linear Regression modelling results using Python: **Age vs Loyalty points**

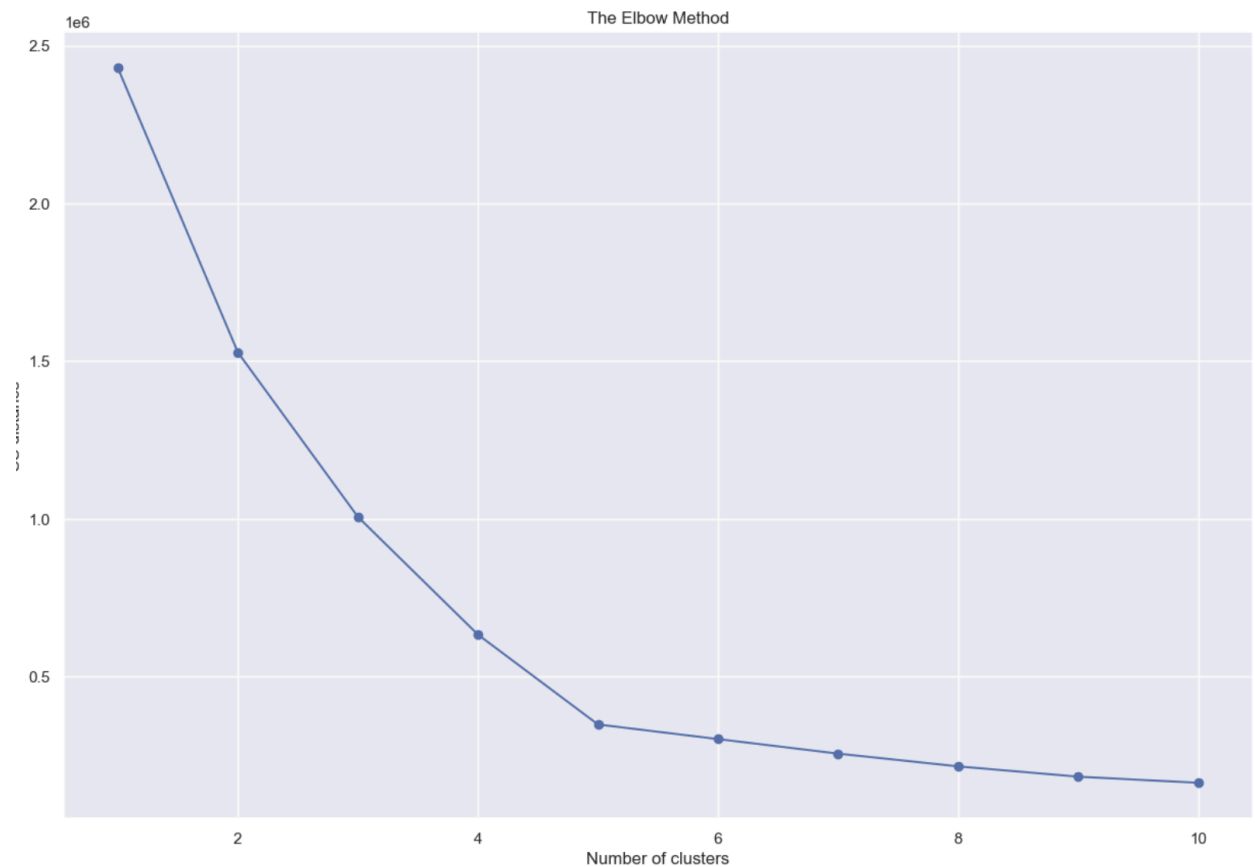


## 1.5 Decision tree modelling

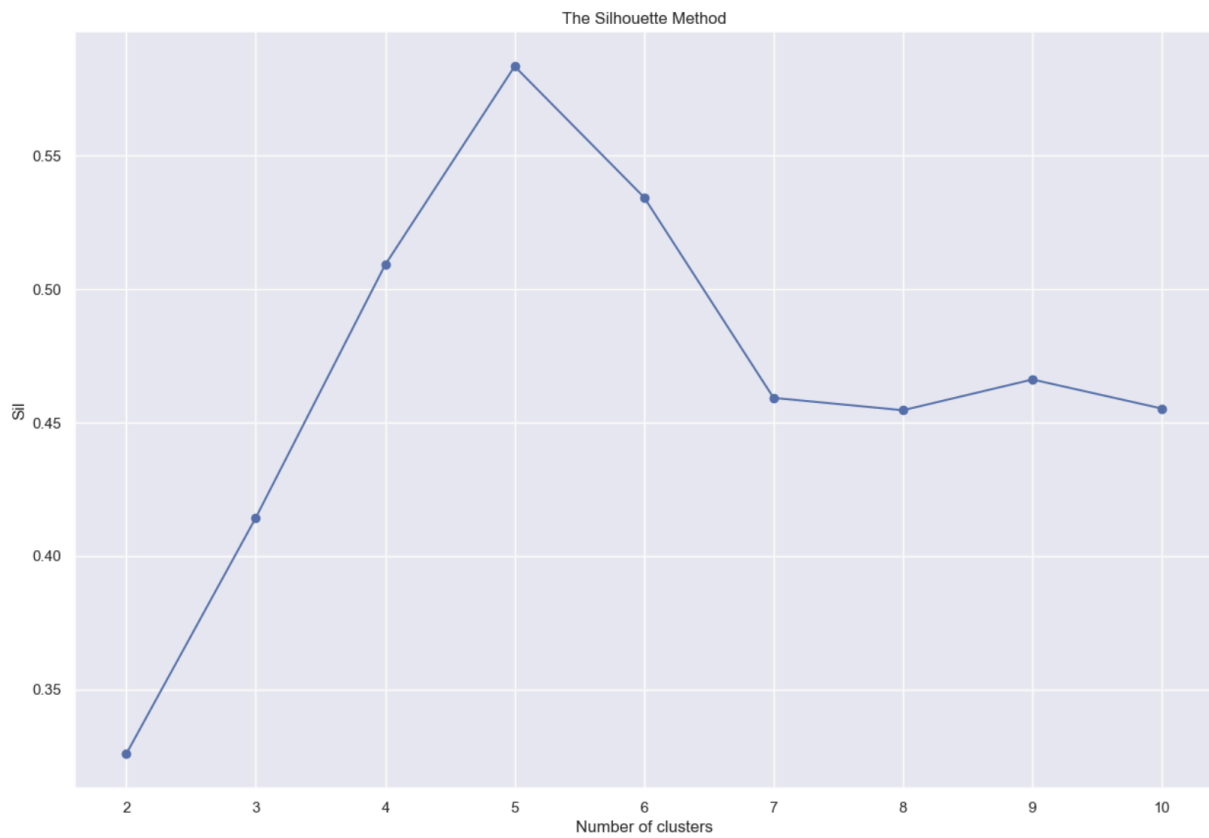


## 1.6 Clustering Using Kmeans

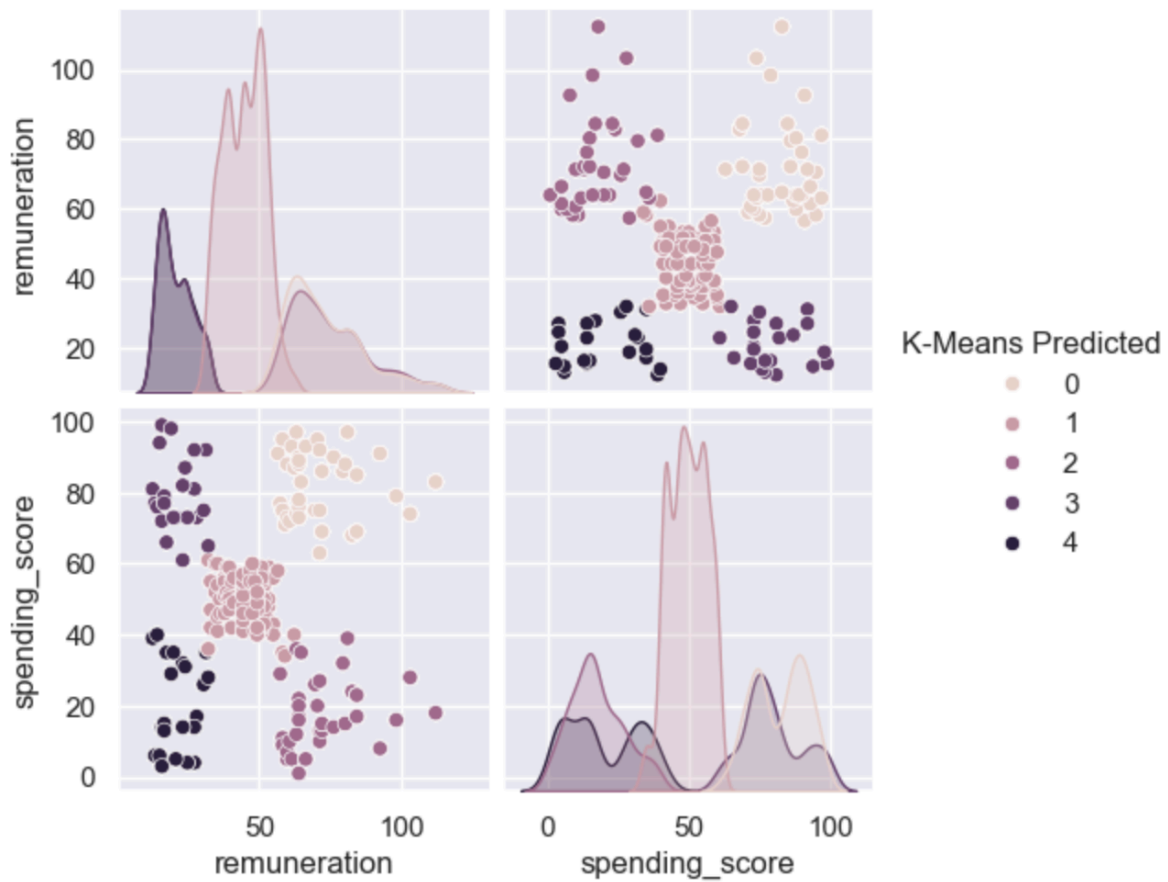
Here the elbow and silhouette method were used to determine the best fit cluster size. In the elbow method we see a clear reduction of inertia at 5. This measures how well each point fits into its own cluster versus others. While the silhouette method peaks at 4 and 8, our pairplot showed that only 4 clusters results in a very spread out group #1. Using 5 clusters captures the lower remuneration/ lower spending score much more precisely, we see clear, tight and distinct groups when using 5 clusters for remuneration vs spending score.



## 1.7 The Silhouette method



## 1.8 The 5 Group Pairplot



```
K-Means Predicted
1    774
0    356
2    330
4    271
3    269
Name: count, dtype: int64
```

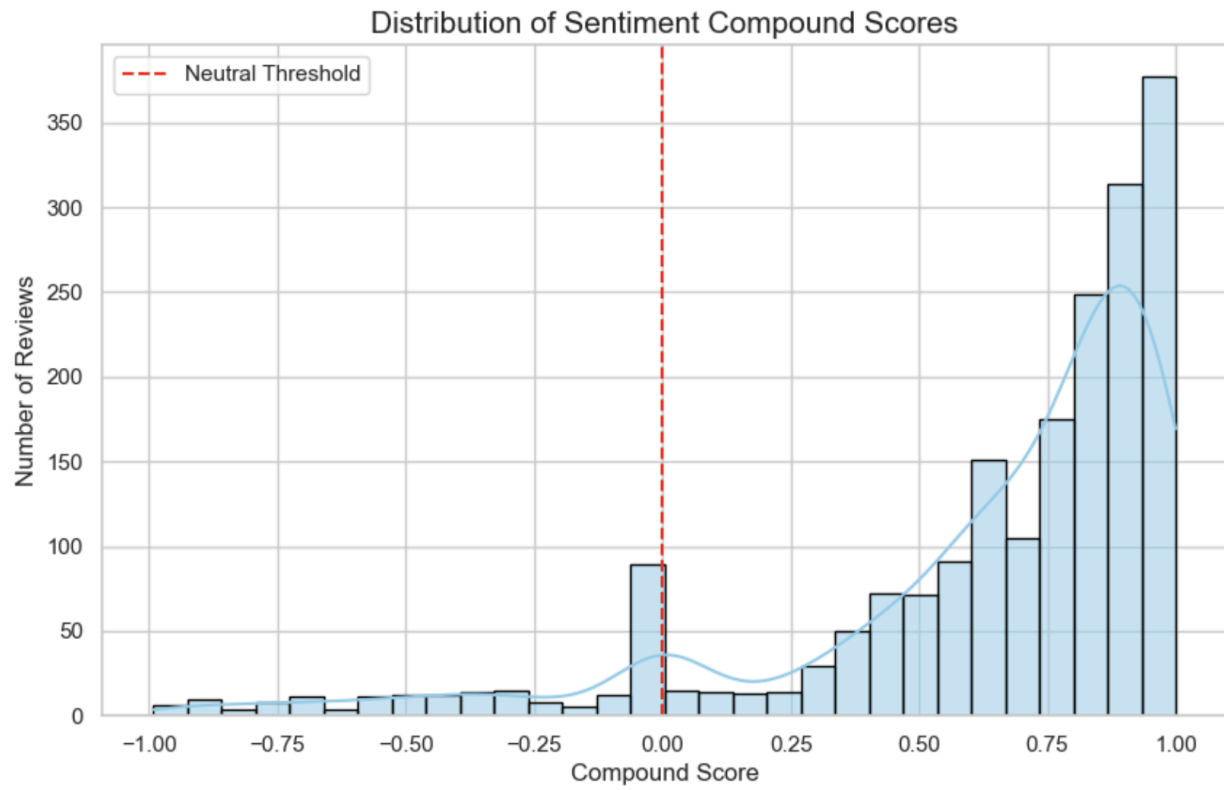
## 1.9 Cluster group breakdown:

Cluster groups are:

- low remuneration + low spending score
- median remuneration + median spending score
- low remuneration + high spending score
- high remuneration + low spending score
- high remuneration + high spending score

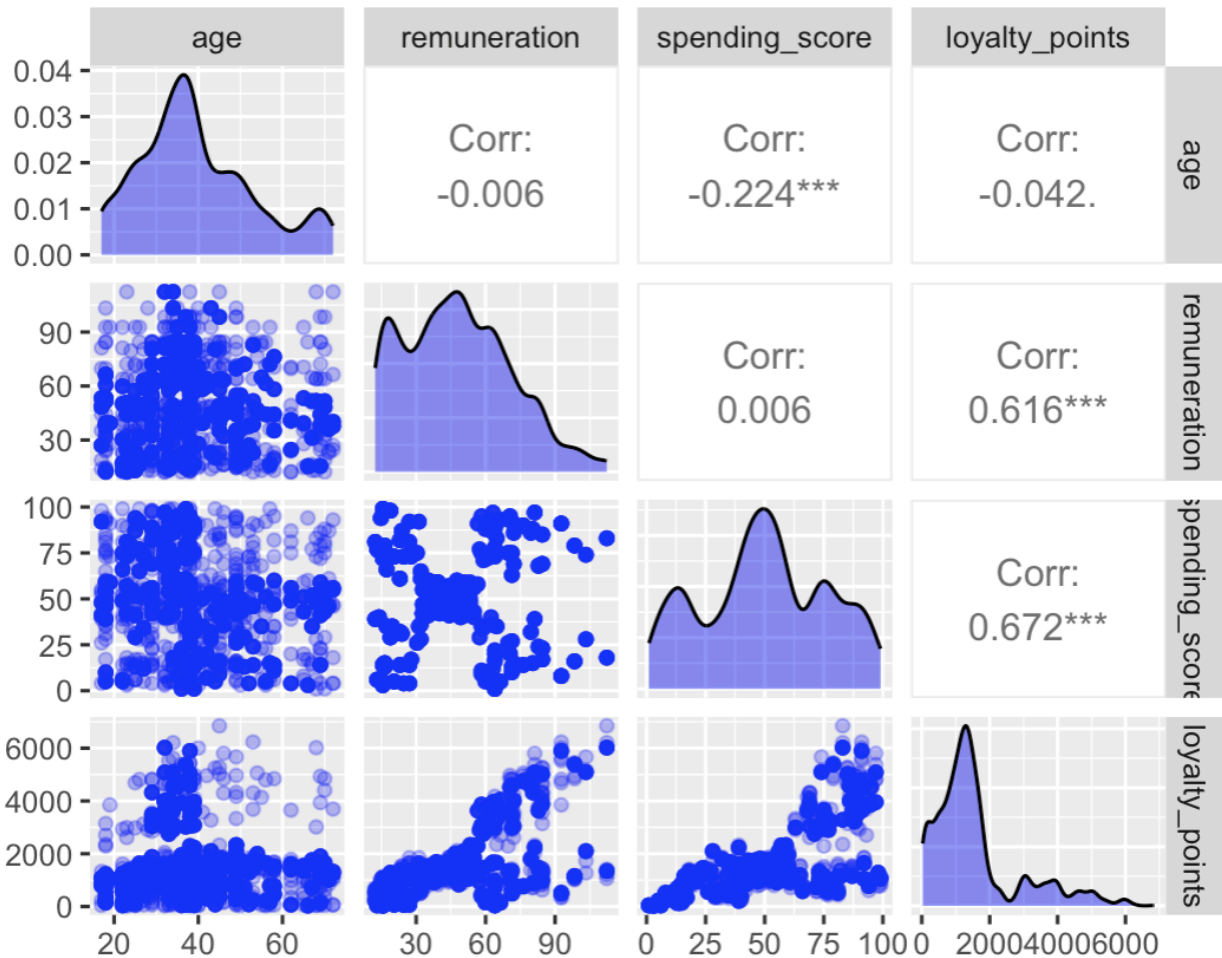
The majority of people (774) fall into this median remuneration + median spending score range whilst the rest are relatively evenly distributed. The smallest group is 269 versus 356 highest.

## 1. 11 Compound review score analysis



## 1.12 Relationship Pairplot

### Relationship Pairplot



## 1.13 Testing Skew, Kurtosis and Normality

```
# Specify the skewness and kurtosis functions.
skewness(turtle$loyalty_points)
kurtosis(turtle$loyalty_points)

# checking for normality
shapiro.test(turtle$age)
shapiro.test(turtle$remuneration)
shapiro.test(turtle$spending_score)
```



```
> skewness(turtle$loyalty_points)
[1] 1.463694
> kurtosis(turtle$loyalty_points)
[1] 4.70883
>
> # checking for normality
> shapiro.test(turtle$age)
```

Shapiro-Wilk normality test

```
data:  turtle$age
W = 0.95242, p-value < 2.2e-16
```

```
> shapiro.test(turtle$remuneration)
```

Shapiro-Wilk normality test

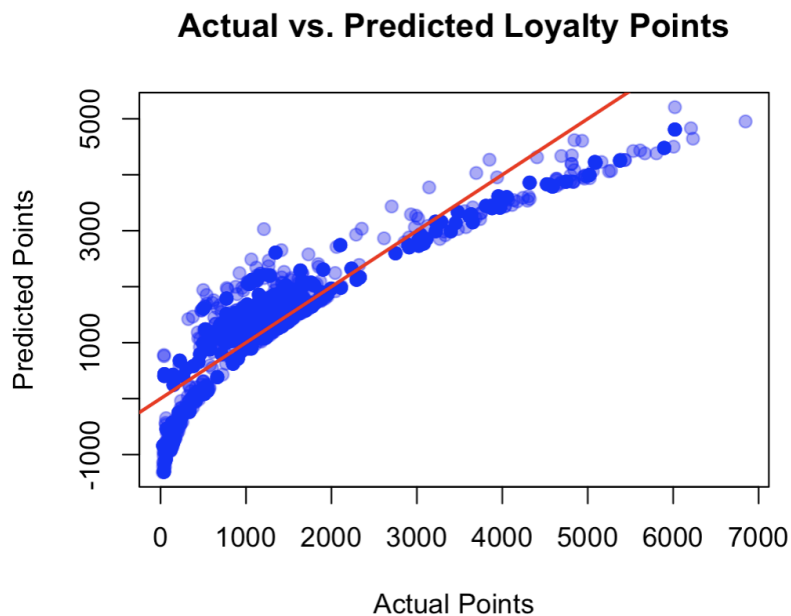
```
data:  turtle$remuneration
W = 0.96768, p-value < 2.2e-16
```

```
> shapiro.test(turtle$spending_score)
```

Shapiro-Wilk normality test

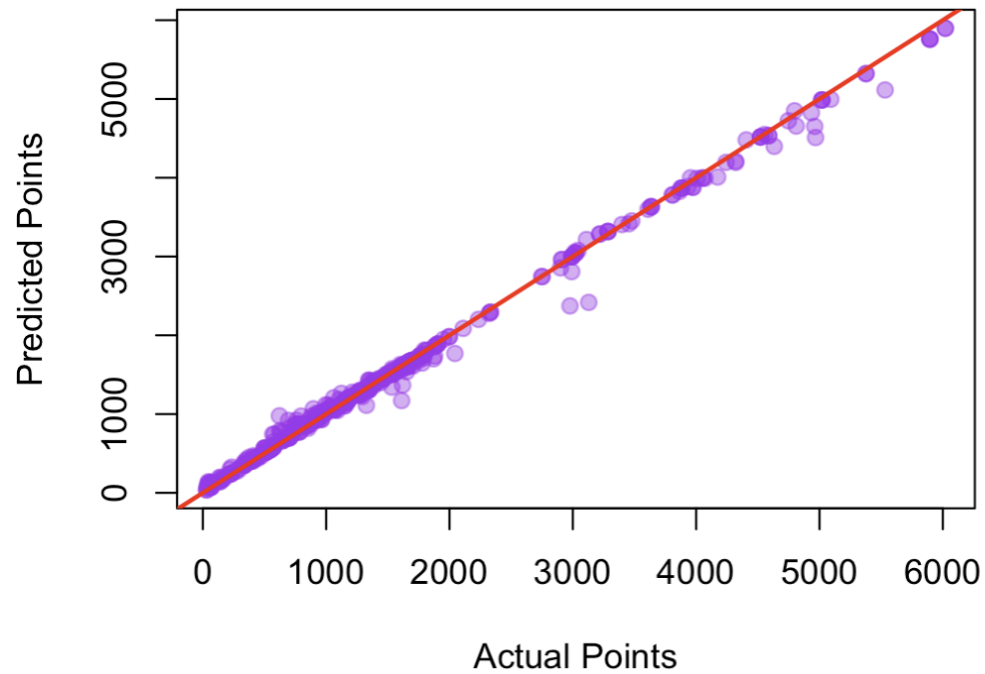
```
data:  turtle$spending_score
W = 0.96835, p-value < 2.2e-16
```

#### 1.14 MLR actual vs predicted results in R



## 1.15 Random Forest Actual vs Predicted Test Set

### Random Forest: Actual vs. Predicted (Test Set)



## 2.1 Age versus spending score

